Rishab Navadgi

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EDUCATION

Bachelor of Science, Materials Science and Engineering **Aug 2025 – Dec 2029**
University of Illinois at Urbana-Champaign

EXPERIENCE

Chassis Team Member | Illini Electric Motorsports, Urbana, IL **Aug 2025 - Present**

- Developed custom ply sweeper to optimize composite layups for 3-point bend testing, utilizing Classical Laminate Theory (CLT) to derive bending stiffness and skin failure, and sandwich beam theory for core shear and skin wrinkling.
- Conducting material trade study for front impact protection structures, comparing carbon fiber laminates against steel and aluminium based on mass, structural equivalency, cost, and manufacturability.
- Owned composites manufacturing for an experimental resin for dry carbon fiber and developed VARTM procedures that reduced cures from hours to under 30 minutes via localized or oven initiation.
- Authored SOPs for the project covering catalyst ratio handling, infusion setup, front initiation, and cure verification.
- Supported manufacturing and layup of carbon fiber sandwich panels, preparing test coupons for tensile, bond strength, and three-point bend testing to validate material performance and manufacturing processes.
- Created mass-tracking spreadsheets that integrated 20+ subsystems, ply schedules, and material properties, improving vehicle weight predictions to within ±2–3%.

Research Assistant | Laboratory for Advanced Space Systems at Illinois, Urbana, IL **Aug 2025 - Present**

- Developing excel based tools to calculate stiffness and preload values for 200+ fasteners for FEA modelling.
- Compiled the research group's fastener modelling reference document, standardizing element selection, stiffness assumptions, and modelling practice so joint analyses are repeatable across the lab.
- Designed a 3D printed satellite model for a University Nanosatellite Program proposal, simplifying model from 500 components to 20 and optimizing print process planning.

Team Captain | Waubonsie Valley Robotics, Naperville IL **Sep 2022 – May 2025**

- Led a team of 8 students to design a robot for Vex Robotics Competition, culminating in 2 State competition qualification, top 10 rankings, and winning awards for excellence in competition and robot construction.
- Developed and tested mechatronic mechanisms unique to our team that achieved competition goals more effectively and enabled real-time performance optimizations.

Student Technician | Waubonsie Valley High School, Aurora, IL **Aug 2023 - May 2025**

- Managed and maintained 5+ fabrication machines (3D printers, laser cutters, CNCs), including routine servicing, troubleshooting, and advising on equipment purchases.

Research Assistant | Pangea Solutions, Palatine, IL **May 2024 - Aug 2024**

- Lead an intern team to design a UAV curriculum and educational materials for students, reaching 200+ students.

PROJECTS

Aero Tracking SR | Skills: CAD/CAM, Rapid Prototyping, 3D Printing **Nov 2024 - May 2025**

- Collaborated with a team to develop a lightweight fixed-wing search and rescue (SR) drone.
- Engineered a fully 3D-printed UAV, achieving final mass of <850 grams and thrust to weight ratio of 1.16.
- Conducted testing of 3D printing materials to evaluate material properties for structural components.
- Performed process optimization studies on Light-Weight PLA, calibrating 5 parameters to ensure consistent print quality and ease of manufacturing.
- Integrated control mechanism using paired elevons to achieve full yaw, pitch, and roll authority, reducing manufacturing complexity and supporting a lightweight, low-cost design.

TECHNICAL SKILLS

Relevant Coursework | Phase and Phase Relations, Introduction to Nanosatellite Design
Software | Creo 10.0, Siemens NX, AutoCAD, NX Nastran, Java, Python
Fabrication and Prototyping | 3D Printing (FDM/SLA), Water Jet, Mill, Lathe, Laser Cutters (CO2/Fiber lasers)